Air University

Computer Networks (Lab)

Year 2025

Mr. Waseem Younas

Academic Year: 2024 – 2028

Department of Cyber Security

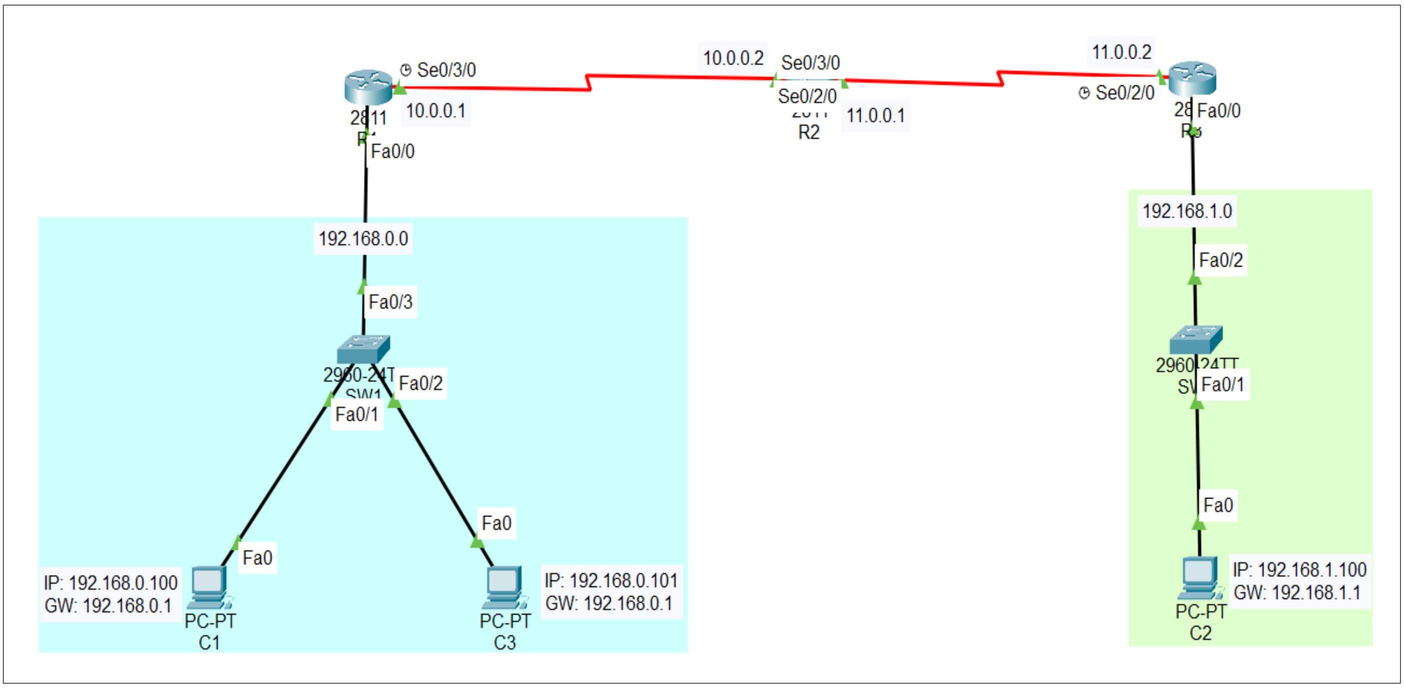
USAMA ARSHAD | 247141

BS – Cyber Security – Fall – 24 (A)

Date of Submission: December 25, 2025

Assignment # 03

Network Diagram



(i) The Final Routing Table from R2

```
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.2/32 is directly connected, Serial0/3/0
     11.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       11.0.0.0/30 is directly connected, Serial0/2/0
L       11.0.0.1/32 is directly connected, Serial0/2/0
R    192.168.0.0/24 [120/1] via 10.0.0.1, 00:00:06, Serial0/3/0
R    192.168.1.0/24 [120/1] via 11.0.0.2, 00:00:14, Serial0/2/0
```

(ii) Show Ip Protocols from R2

```
R2#show ip protocols
Routing Protocol is "rip"
Sending updates every 30 seconds, next due in 4 seconds
Invalid after 180 seconds, hold down 180, flushed after 240
Outgoing update filter list for all interfaces is not set
Incoming update filter list for all interfaces is not set
Redistributing: rip
Default version control: send version 2, receive 2
  Interface             Send  Recv  Triggered RIP  Key-chain
  Serial0/3/0           22
  Serial0/2/0           22
Automatic network summarization is not in effect
Maximum path: 4
Routing for Networks:
        10.0.0.0
        11.0.0.0
Passive Interface(s):
Routing Information Sources:
        Gateway         Distance      Last Update
        10.0.0.1             120      00:00:00
        11.0.0.2             120      00:00:09
Distance: (default is 120)
R2#
```

(iii) One Successful Ping from C1 To C2

C1

Physical Config Desktop Programming Attributes

Command Prompt X

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Request timed out.
Reply from 192.168.1.100: bytes=32 time=12ms TTL=125
Reply from 192.168.1.100: bytes=32 time=10ms TTL=125
Reply from 192.168.1.100: bytes=32 time=12ms TTL=125

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 10ms, Maximum = 12ms, Average = 11ms

C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time=40ms TTL=125
Reply from 192.168.1.100: bytes=32 time=11ms TTL=125
Reply from 192.168.1.100: bytes=32 time=11ms TTL=125
Reply from 192.168.1.100: bytes=32 time=18ms TTL=125

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 11ms, Maximum = 40ms, Average = 20ms
```

(iv) Traceroute Output from C1 To C2

```
C:\>tracert 192.168.1.100

Tracing route to 192.168.1.100 over a maximum of 30 hops:

  1    0 ms      0 ms      1 ms      192.168.0.1
  2    1 ms      3 ms      1 ms      10.0.0.2
  3    1 ms      3 ms      11 ms     11.0.0.2
  4    12 ms     11 ms     12 ms     192.168.1.100

Trace complete.
```

THE END